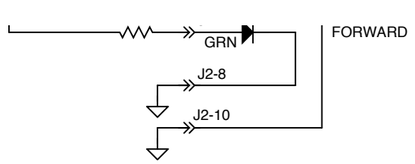
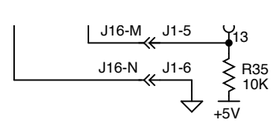
00-09 Hall Effect Schematic Modified 5-5-09 J1-N to J11-A

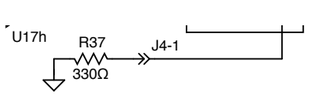
Added varistors across K18, D10, and Magnet 7-29-09 D.G.Varistors had been removed from K18.

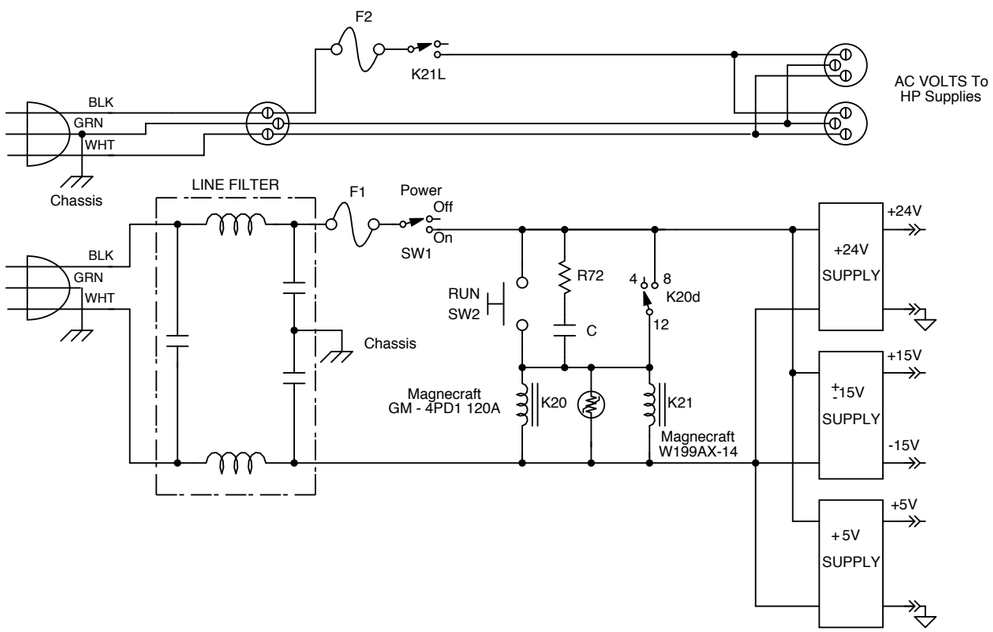
R77 1.3K
R1 10K
R2 10K
OP497 U1-A
R3 102K
R4 1.5K
0V to -144mV
J1-E
J1-F
J15-7
J15-8
HP6274B MASTER
A5 A6 A7 A2 A10 A8 A3 +S -S A9
J15-2
J15-3
J15-4
J15-1
J1-1
0V-10V
CONNECTOR BLOCK
DAC 0 OUT
DGND
AGND
A CH.0 A CH.1
J1-3
J1-2
J1-A
J1-B
R73 10K
R74 10K
In Breakout Box
6W R5 18K
HP6274B SLAVE
A1 A2 A3 +S A10 A9 -S A8 A5 A6 A7
D10 1N1184A
-15V
R77 1.5MΩ
Offset P3 Adjust 200Ω
R6 1K
C35 79nF (10Hz Filter)
R7 200K
OP497 U1-B
INA 117 U2
SHUNT 10mV / A
J1-8
J1-7
K19R
K21R
K19L
K18a
K18b
R75 1KΩ 100W
To J15-4
J15-9
J15-10
MAGNET SPECS 350V DC @4A 74Ω
120V / 74Ω = 1.62A
Set P1 for 3.2A through shunt
3.2A x (10mV) = 32mV / A
32 mV x (200)=6.4V
+15V C11 .1μfd
U5 AD584
10V
C12 1000pf
P1 100K
U1-C
OP497 Adjust for 6.4V
R8 1K
R9 100K
C7
U3 LM311
C8
R10 2K
CURRENT OVERLOAD
+5V
D8 1N4448
R16 1MΩ
C13 1μF
U6b U6c
POR
PC MAGNET OVERLOAD DATA BIT PA4
High to Low Transition
J1-13
R18 100Ω
R17 100K
74HCT14 U7a
U8b
Q8 IRFZ44N n-Channel Enhancement
Q9 IRFZ44N n-Channel Enhancement
J1-N
J11-A
LINE 2
"BOX ON" PB3
Magnet Shunted (To Computer)
J1-H PB2
R13 100K
R11 100K
C9
R76 1K
On back of board
150mV
U4 LM311
R12 1.5K
C10
R14 1K
R15 330Ω
J2-6
J2-5
Yellow Field≤5mA
MAGNETIC FIELD OFF
SPARE PA6
SPARE PA7
J1-15
J1-16
R20 100Ω
R19 100K
R22 100Ω
R21 100K
74HCT14 U7b U7c
74HC20 U9a
ULN2003A U10a
R23 4.64k
R24 1K
Q7 2N2905A
+24V
J9-7 J10-7
J14-2
VR1 1N4742A 12V
Magnecraft W199X-13
K19
D7 1N4007
MAGNET SHUNTED
Blinking red LED #604-L56BSRDB
J13-2
J14-1
J13-1
74HC04 U13a
PC ALARM DATA BIT PA5
J1-14
R26 100Ω
R25 100K
U7d
MAGNET CURRENT ZERO
MAGNET CURRENT DIRECTION
Data Bit PA3
J1-12
R39 100Ω
R38 100K
74HCT14 U7e
C21 470pF
R40 10K
U12a
U12b
74HC221
C22 470pF
R41 10K
74HC02 U14a
GATE
74HC08 U15a
U6d
74HC240A U17a
U17b
R44 330Ω
R45 330Ω
J2-3 RED
J2-1 GRN
J2-2
J2-4
REVERSE
FORWARD
ULN2003A U10b
J9-6
J10-6
R42 4.64k
R43 1K
Q6 2N2905A
J14-4
K18
D6 1N4007
Potter & Brumfield KRP11DG
J14-3
+24V
74HC74 U16a
SAMPLE CURRENT DIRECTION
Data Bit PA2
J1-11
R47 100Ω
R46 100K
74HCT14 U7f
74HC04 U13b
U11b
J9-5
J10-5
R48 430Ω
R49 1K
Q5 2N2905A
K17
D5 1N4448
Magnecraft W172DIP-21
74HC240A U17c
U17d
R50 330Ω
R51 330Ω
J2-9 RED
J2-7
REVERSE
R27 1K
R29 20K
R28 1M
R75 2K
OP497 U1-D
R30 1K
R32 332K
C29
U18 LM311
C30
R33 2K
R31 23.3K
P2 10K
Adjust for 115.5 mV
VACUUM GAUGE
(+)
(-)
J1-17
J1-18
5.5mV
C28 1μF
AC Ground was attached at this point from the secondary of the TC Vacuum Gauge. It was attached to chassis ground inside the box near the frontmost DC supply. It has been disconnected and covered with heat shrink tubing. The cryostat also receives AC grounding through the water pipes attached to it. DKG 6-3-09.
U9b
ULN2003A U11a
ALARM
K20c
VACUUM INTERLOCK
512-MV5437
74HC240A U17e
U17f
J3-3 GRN
J3-2 RED
J3-1
R36 330Ω
VACUUM OK DATA BIT PB0
J1-C
WATER OK DATA BIT PB1
J1-D
WATER INTERLOCK
J16-L
J1-4
R34 10K
74HC240A U17g
U16b 74HC74
J4-3 GRN
J4-2 RED

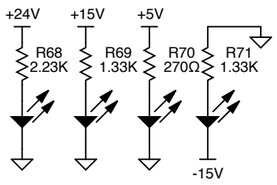
00-09 Hall Effect Schematic

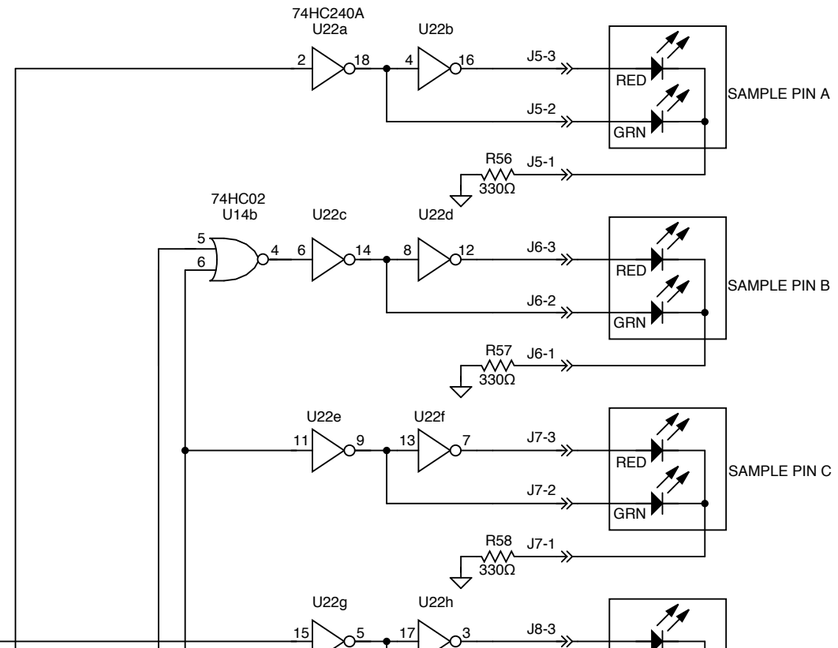


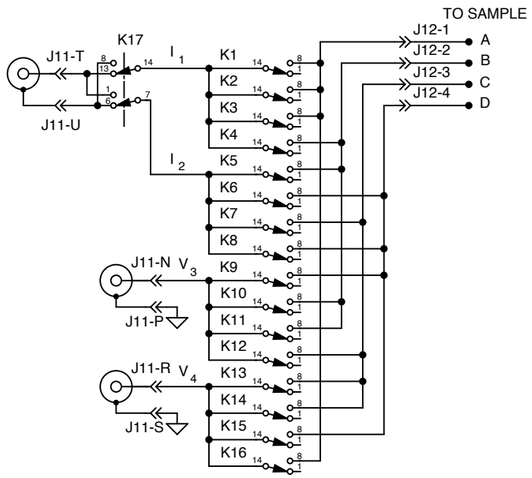


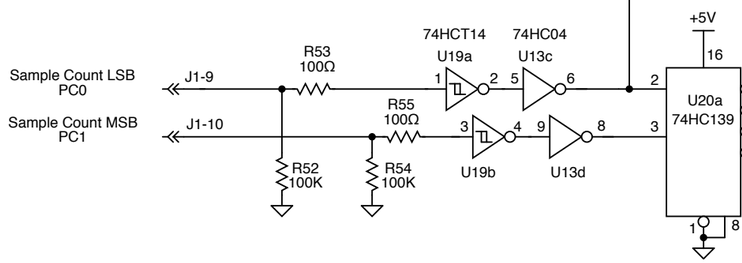


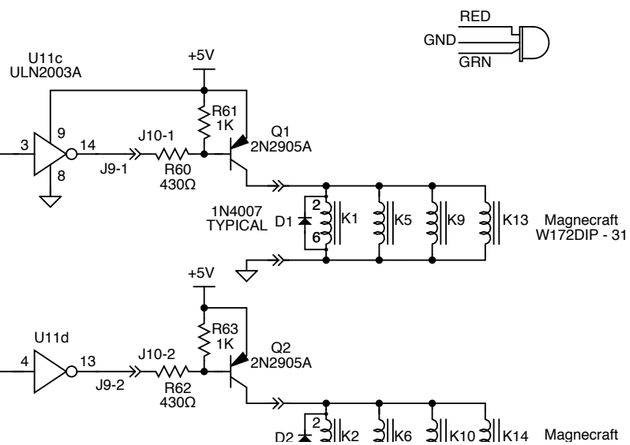


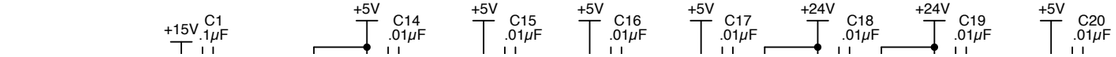


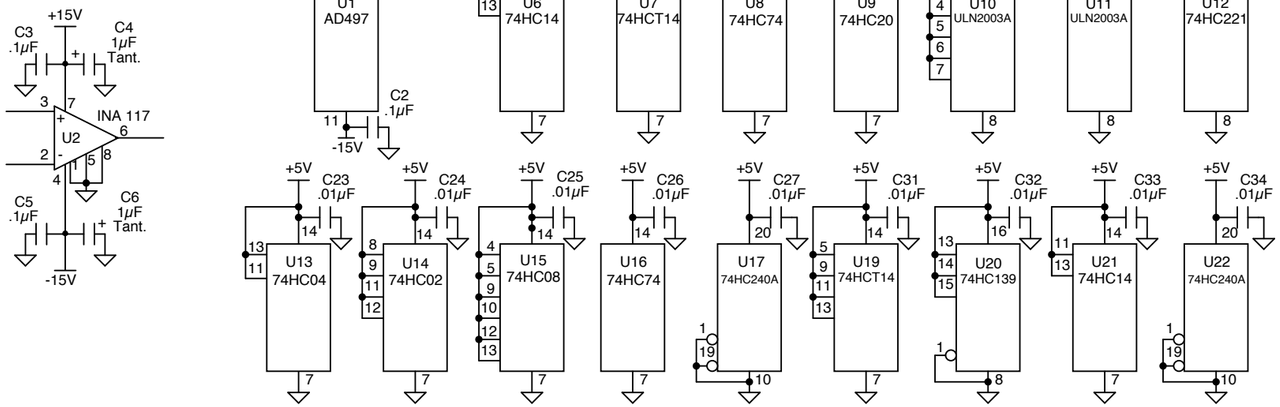


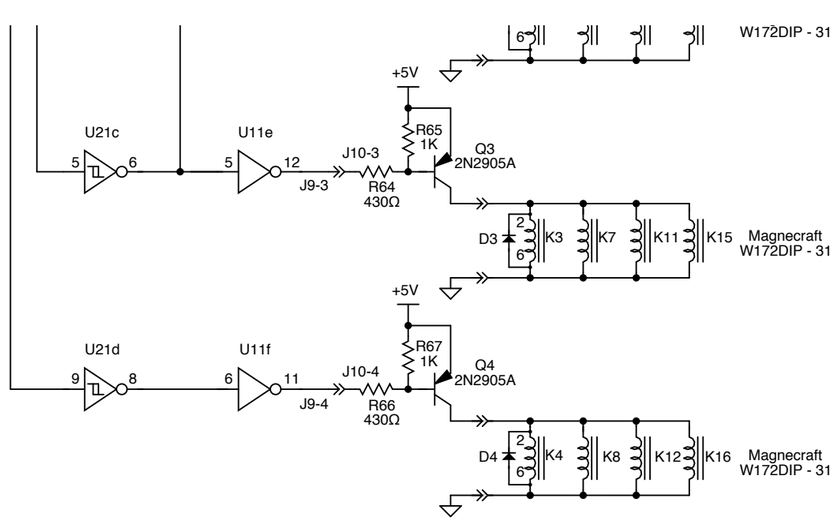












PA2 1 = Reverse 0 = Forward

